THE GASEOUS METABOLISM OF THE BRAIN OF THE MONKEY[1]

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Received for publication September 20, 1944

The amount of oxygen used by the living brain, per unit of weight and time[2] and under varying conditions of activity, remains a matter for speculation in spite of considerable experimental study. This can be seen from the following list of representative values for the Q_{O_2} of brain found in the literature:

		Q_{O_2}
	In vivo	
Alexander et al. (1)	Dog	39
Handley et al. (9)	Dog (perfusion)	32.7
Gayda (8)	Dog	30.0
Rein (17)	Dog	30.0
Schmidt (18)	Dog (perfusion)	22.5
Hou (10)	Dog	16.8–23.4
Yamakita (24)	Rabbit	28.2
Chute and Smyth (4)	Cat (perfusion)	9.9–15.0
	In vitro	
Warburg (22)	Rat cortex slice	10.7
Loebel (12)	Rat cortex slice	11.4–13.7
Dickens et al. (5)	Rat cortex slice	12.4
Jowett et al. (11)	Guinea pig cortex slice	11.8
Quastel et al. (16)	Guinea pig cortex slice	14–15
Elliott et al. (7)	Rat gray matter homogenate	15.5
Elliott et al. (7)	Rat whole brain homogenate	11.1

The differences within the *in vivo* group are so great as to cast doubt upon the validity of some of the figures, but whether one should prefer the largest (on the general ground that the necessary experimental abnormalities would tend to reduce cerebral metabolism) or the smallest (because of the strong probability that not all of the blood flow measured as cerebral and used in the calculation of Q_{O_2} actually went to the brain) is utterly unknown. The variations within the *in vitro* group are smaller, but the differences between these and the *in vivo* values leave doubts as to whether the consistency of the *in vitro* figures indicates that a true physiological constant is being dealt with, or whether the artificial conditions

[1] The work described in this paper was done under a contract, recommended by the Committee on Medical Research, between the Office of Scientific Research and Development and the University of Pennsylvania. Financial support was also received from the National Committee for Mental Hygiene from funds granted by the Committee on Research in Dementia Precox founded by the Supreme Council, 33° Scottish Rite, Northern Masonic Jurisdiction, U. S. A. Permission for publication has been granted.

[2] The expression Q_{O_2} has come into general use for this purpose. It is defined (14) as μl. of O_2 used per milligram of dry tissue per hour. Obviously the quantity would be the same if milliliter per gram of tissue were used instead, but it is important to note that the Q_{O_2} is referred to dry weight (which in the brain is about 20 per cent of the wet weight) and that the unit of time is specified as one hour.

of the *in vitro* experiment lead to abolition of a large proportion of the normal metabolic activity and leave only a relatively rugged moiety to be called the Q_{O_2}.

These questions had a certain amount of practical importance in relation to neurology and psychiatry during peace time, but the current emergency has greatly increased the need for more precise information about them. This is because the functional capacity of the human brain is definitely the limiting factor in the activities of implements of warfare such as airplanes, tanks, submarines and certain other terrestrial and aquatic vehicles, as it is also the critical factor in survival from shock, hemorrhage and anoxia due to disease, high altitude, or the effects of noxious gases. Therefore more precise information about the metabolism of the living brain *in vivo* offers some prospect of improving, not only our understanding of the vulnerability of the brain to stresses such as these, but also the procedures intended to compensate for the physiological limitations thus imposed.

In vivo estimations of the metabolism of the brain, as of any living tissue, involve collection and analysis of representative samples of venous and arterial blood, and measurement of the volume of blood flow corresponding with these. The latter, which was the stumbling block in the past, can be circumvented by the use of an animal (the monkey) in which the anatomical relations of the cerebral circulation are favorable for such measurements, and a measuring device that is adequately dependable and accurate. This has already been done (6). The only addition required for estimations of cerebral metabolism is provision for the collection of samples of cerebral venous and of arterial blood and this has been done in the experiments about to be described.

METHODS. Except for 3 spider monkeys (*ateles pentadactylus*) weighing 5–5.8 kilos, the experiments were made on Indian rhesus monkeys ranging in weight from 2.8 to 4.6 kilos. They were lightly anesthetized by intraperitoneal (intramuscular in the spider monkeys) injection of nembutal in dosage of about 20 mgm. per kilo; this was supplemented by intravenous injection of 5 mgm. (total—not per kilo) doses of pentothal sodium as required. The operative procedures were the same as in the earlier experiments (6) with only two important differences: the basilar artery was exposed through a trephined opening, the button of bone being replaced and sealed with bone wax after the artery was tied, and both internal jugular veins were exposed and their major branches ligated as far back toward the skull as possible. A flexible plastic catheter having an outside diameter of 2 mm. and a bore of 1 mm. was passed up each internal jugular as far as it would go and tied in place; the two catheters were joined by a glass Y-connection for the collection of cerebral venous blood; the veins were also ligated on the cardiac side to prevent reflux bleeding. A tracheal cannula was inserted and was connected, through a suitable cannister containing soda lime, to a counterbalanced recording spirometer having a total capacity of 500 cc., usually with a set of water valves in the circuit to humidify the inspired gas, minimize rebreathing, and facilitate the use of other gas mixtures if desired; this closed respiratory system was filled with oxygen, flushed at suitable intervals to remove exhaled nitrogen, and used to record both the total oxygen consumption and the depth and rate of the respirations. In case of respiratory failure the pulmonary gas exchange was cared for by intratracheal insufflation of oxygen through a small rubber catheter. Systemic blood pressure was measured from a femoral artery by means of a mercury manometer. Intravenous injections were made through a burette-cannula system connected with a femoral vein. Coagulation of the circulating blood was prevented by intravenous injection of a single dose of Pontamine

Fast Pink BL (100 mgm. per kilo) supplemented by 1000 unit doses of heparin at hourly intervals.

The measurement of cerebral blood flow was accomplished, as in our earlier experiments (6), by means of a bubble flow meter of about 6 ml. capacity intercalated in the stream of blood passing through both common carotid arteries, the external carotid and basilar arteries being tied. Arterial blood samples were usually collected from the internal carotid end of the flow meter, occasionally from a femoral artery. Several attempts were made to measure the basilar blood flow as in our previous experiments but these were unsuccessful because the rhesus monkeys used in the present work were smaller and less tolerant of the operative procedures than those used before, and the spider monkeys turned out to have very small basilar arteries.

The course of a complete experiment was as follows: The intravenous injection system was installed (with the aid of 5–10 mgm. doses of pentothal sodium intraperitoneally or procaine locally) so that the otherwise insufficient nembutal narcosis could be supplemented by intravenous pentothal as required. These and the subsequent operations were accomplished with all possible precautions against bleeding, a high frequency electric or a simple heated iron cautery being used liberally, and all grossly visible blood vessels being tied. The internal jugular and carotid vessels were exposed on both sides. All accessible branches of the former cephalad to the point of insertion of the catheters were tied and ligatures were passed around the external carotids on the cardiac side of the origin of any branches. The tracheal and femoral arterial cannulae were then inserted and records of blood pressure, oxygen consumption and pulmonary ventilation were begun. The internal jugular catheters (previously filled with heparin solution) were inserted and tied in place. A cannula was introduced into the other femoral artery for the collection of arterial blood. Then a pair of samples of blood (cerebral venous and arterial) was collected to provide orientation as to the cerebral A-V oxygen difference under conditions as close to normal as possible. (This was not done in all experiments.) The blood withdrawn was usually replaced by an equal volume of blood saved from the preceding monkey, or of isotonic gelatin in saline. Then the operation for exposure of the basilar artery (the most formidable part of the preparation) was carried out; in some cases the artery was tied at once, in others the ligature was passed but not tied until the carotid cannulae had been inserted, decision on this being based on the presence or absence of arterial bleeding from rupture of small branches when the ligature was passed. The anticoagulants were then injected, the external carotids tied, and installation of the carotid cannulae carried out in such a way that at least one internal carotid was always carrying blood; flow was established through both carotids before the meter was connected. The meter was filled with gelatin solution from a by-pass to the femoral venous injecting system and then blood was admitted from the carotid cannulae, displacing gelatin and carrying any air bubbles present into the femoral vein. Flow through the meter was then established by clamping the bypass to the femoral vein, and a previously patent U-connection between the pairs of cannulae, as soon as a steady state was attained. The blood samples for the control period were collected, a measurement of blood flow being made during the collection of venous blood. The remainder of the experiment was then proceeded with. Depending on circumstances, this involved collection of blood samples (with simultaneous measurement of blood flow) during spontaneous changes, following withdrawal and reinjection of varying amounts of blood, infusions of additional amounts of blood or gelatin solution or of epinephrine in gelatin (to increase cerebral blood flow), and the intravenous or intracarotid injection of convulsant doses of metrazol, picrotoxin and nikethamide, of depressant doses of pentothal, and a few other agents.

The blood samples, which measured about 5 ml. each, were collected over mercury in small tonometers, chilled at once, and analyzed for CO_2 and O_2 content within 8 hours by the manometeric method; all analyses were done in duplicate and the usual precautions were taken (14).

At the end of each experiment the animal was exsanguinated and an injection of India

ink was made through the internal carotids. The distribution of the ink, in the brain and elsewhere, was checked by dissection, as was also the position of the jugular catheters. The brain was then removed and weighed anterior to the level of the basilar ligature, this factor being used in calculating blood flow and metabolism per unit weight of tissue. The results of these terminal injections confirmed those of the earlier experiments: the only extracerebral tissues injected were the eyes, eyelids and orbital tissues, with occasional small patches in the temporal and occipital musculature. The tips of the venous catheters either were just outside the skull and clearly visible from the inside, or had entered the cranial cavity and turned backward (usually) or forward into the cerebral sinus system. Collection of venous blood was made from both sides because of the fact (3) that one jugular may drain the entire cortex, the other the ventricular and choroid plexus system; a sample of blood drawn from one jugular bulb, or even from the cerebral sinuses draining into it, therefore cannot be counted upon to be truly representative of cerebral venous blood.

A total of 32 experiments were made. Of these 6 had to be discarded either because of premature death of the animal or technical imperfections such as a patent vertebral artery.

RESULTS. 1. *The "normal" metabolic rate of the brain.* The results of our 11 best experiments are summarized in table 1. These are selected on the following basis: *a*, cerebral A-V oxygen differences of the same order as those found before extensive operation and ligation of blood vessels; *b*, active cerebral functions, evidenced in brisk ocular reflexes, respiration and total oxygen consumption similar to those seen early in the experiment, and occasional spontaneous muscular movements. We believe these to be as close to the metabolic activity of the living brain *in vivo* as it is possible to get by a method involving as extensive dissection as this. The figures for O_2 consumption can be converted to Q_{O_2} (μl. per mgm. dry tissue per hour) by multiplying them by three[3].

2. *Variations in cerebral metabolism.* All of our findings are presented in figure 1 as a smoothed frequency graph showing the distribution of different levels of cerebral O_2 consumption in relation to different levels of cerebral functional activity. The tendency of cerebral O_2 uptake to run parallel with cerebral functional activity is quite evident. There is some overlapping but this is not surprising in view of the arbitrary basis for the classification into these three categories. The differences among the three mean values are statistically significant.

The lowest figure for cerebral O_2 consumption (per 100 grams of fresh brain per minute) was 0.39 ml. in a moribund animal without any signs of cerebral activity. The highest was 6.5 ml. during a pricrotoxin convulsion (table 4, expt. 27). The lowest compatible with subsequent recovery of cerebral activity was 1.85 ml. following hemorrhage (table 2, expt. 26). The "physiological" range of cerebral O_2 uptake under these experimental conditions therefore was from nearly double to about half the resting "normal" value. (See table 6, p. 44.)

3. *Effects of intentional decrease or increase in cerebral blood flow.* (a) *Decrease.*

[3] This is derived as follows, taking the average O_2 consumption as an example: (1) Divide by 100 to reduce to milliliter per gram (or μl. per mgm.) per minute (= 0.037). (2) Multiply by 5 to convert to dry weight (= 0.185). (3) Multiply by 60 to convert to O_2 uptake per hour (= 11.1). Dividing by 100 and then multiplying by 300 gives the same result as simply multiplying by 3.

This was brought about by removing appropriate amounts of blood. As usual in bleeding experiments, subsequent reinjection of the blood in some cases brought about complete and (for our purposes) permanent restoration of circulation, respiration and ocular reflexes, while in others the improvement was only partial and temporary. Cerebral blood flow was invariably decreased by the

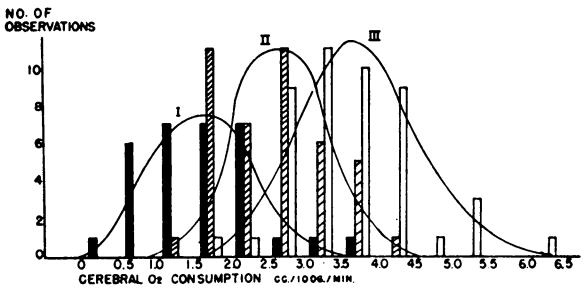


Fig. 1. Relation of cerebral O_2 consumption to cerebral functional activity, arranged in three groups: I spontaneous breathing, ocular reflexes and all other signs of cerebral activity entirely absent (black rectangles indicate observations in this group); II spontaneous breathing of some type present but ocular reflexes and muscular movements absent (indicated by cross-hatched rectangles); III breathing normal, ocular reflexes and occasional movements of extremities present (indicated by open rectangles). The mean values and their standard errors are 1.7 ± 0.15, 2.6 ± 0.13 and 3.7 ± 0.16 for groups I, II, and III respectively.

TABLE 1

EXPT. NO.	GENERAL CONDITION	BLOOD PRESSURE	CEREBRAL				O_2 CONSUMED		CEREBRAL R.Q.	WEIGHT	
			A-V diff.		Blood flow	O_2 uptake	Total	% by brain		Animal	Brain
			O_2	CO_2							
			vols. %		*ml./100g./min.*		*ml./min.*			*kg.*	*grams*
5	Good—light anesthesia	135	7.0	6.0	53.4	3.7			0.85	2.8	90
9	Good—deep anesthesia	84	6.6	6.9	38.0	2.5			1.05	3.2	90
13	Good—light anesthesia	114	8.7	7.3	33.5	2.9	29	9.0	0.84	3.5	88
14	Excellent—active movements	100	10.1	10.2	33.4	3.4	30	11.7	1.01	3.5	114
22	Excellent—active movements	134	7.8	8.0	41.0	3.1	40	6.9	1.02	4.1	89
24	Excellent—reflexes very active	124	6.0	6.0	73.6	4.4	31	12.1	1.00	3.7	85
25	Excellent—reflexes very active	150	9.7	8.2	46.6	4.5	41	8.8	0.85	4.1	80
26	Excellent—reflexes very active	116	8.2	7.6	40.4	3.3	31	8.7	0.92	3.3	82
27	Excellent—reflexes very active	102	8.8	10.7	49.7	4.4	35	9.4	1.22	3.0	75
30	Excellent—reflexes very active	110	7.8	7.5	53.0	4.1	37	12.3	0.96	5.8	105
31	Excellent—reflexes very active	100	8.2	8.1	53.7	4.4	48	10.0	0.99	5.3	108
Average		115	8.0	8.0	46.9	3.7	36	10.0	1.00	3.8	91

hemorrhage and increased by the subsequent transfusion; the A-V oxygen difference underwent changes exactly the opposite of these. In the cases of complete recovery after transfusion cerebral O_2 uptake came back to or above its control level while in the others it did not. Representative examples of these responses are shown in table 2. The data suggest an inverse rather than a direct relationship between the intensity of cerebral metabolism and the ability of the

brain to withstand the anoxia produced by hemorrhage, but the observations are too few to warrant any categorical statements at the present time.

(b) *Increase.* This was brought about by intravenous infusion of epinephrine (1:20,000 or 1:100,000 in isotonic gelatin) in two spider monkeys in excellent condition, also in one spider and four rhesus animals as a restorative measure after breathing had ceased and reflexes had disappeared. The effects on cerebral blood flow and A-V oxygen difference were uniformly the reverse of those produced by hemorrhage. Cerebral O_2 uptake, however, behaved differently according to the state of cerebral activity before the epinephrine was given. If the

TABLE 2

EXPT. NO.	EXPERIMENTAL CONDITIONS	BLOOD PRESSURE	O_2 CONTENT		CO_2 CONTENT		CEREBRAL			O_2 CONSUMED		CONDITION OF ANIMAL	CORNEAL REFLEX
			Art.	Vein	Art.	Vein	AV-O_2	Blood flow	O_2 uptake	Total	% by brain		
		mm. Hg	*vols. %*		*vols. %*		*vols. %*	*ml./100 g./min.*		*ml./ min.*			
26	Control	116	14.4	6.1	45.6	53.2	8.2	40.4	3.30	31	8.7	Excellent	++
	Bleeding 10 ml.	88	14.5	5.3	45.5	55.2	9.3	33.9	3.14	28	9.2	V. good	++
	Bleeding 10 ml.	70	14.9	3.3	43.9	56.3	11.7	27.9	3.26	28	9.6	V. good	++
	Bleeding 10 ml.	58	14.8	0.8	41.2	57.4	14.0	20.2	2.83	25	9.3	Fair	±
	Bleeding 10 ml.	44	14.7	0.6	35.8	55.8	14.1	13.2	1.85	21	7.2	Bad	0
	Blood returned	114	11.6	3.4	46.0	53.7	8.1	46.0	3.72	34	9.0	Good	+
	15 min. later	110	12.9	3.4	47.4	57.3	9.5	41.4	3.94	35	9.2	Good	+
24	Control	124	11.6	5.6	50.9	56.9	6.0	73.6	4.43	31	12.2	Excellent	+++
	Bleeding 10 ml.	80	12.7	3.4	53.5	61.3	9.3	39.6	3.69	28	11.1	Fair	+
	Bleeding 6 ml.	64	12.5	2.2	54.6	64.4	10.3	30.2	3.12	27.5	9.8	Bad	0
	Blood returned	105	12.2	3.2	51.9	60.3	9.0	32.2	2.92	32	7.8	Bad	0
	Gelatin 10 cc.	138	11.4	5.1	51.5	57.5	6.3	56.5	3.55	34	8.8	Good	++
30	Control	110	12.5	4.7	52.1	59.6	7.8	53.0	4.13	35	12.3	Excellent	++++
	Bleeding 75 ml.	56	10.3	1.5	49.7	59.6	8.8	32.9	3.04	27	11.1	Good	++
	Bleeding 35 ml.	40	9.4	0.9	45.7	54.8	8.5	24.6	2.09	12	18.3	Bad	±
	Blood returned	100	10.0	5.3	45.1	49.1	4.7	62.2	2.93	35	8.9	Good	++
25	Control	150	14.6	4.9	45.9	54.1	9.7	46.6	4.52	41	8.8	Excellent	+++
	Bleeding 10 ml.	100	15.2	2.0	46.7	58.3	13.2	29.5	3.90	32	9.7	Fair	+
	3 min. later	96	13.7	1.3	51.6	59.6	12.4	28.0	3.48	30	9.0	Bad	0
	Transfused 25 ml.	114	13.2	3.3	57.5	65.4	9.9	31.2	3.10	art.	resp.	Bad	0
	30 min. later	56	8.1	3.2	79.6	83.0	4.9	11.2	0.56	art.	resp.	Moribund	0

latter was already optimal (normal breathing, brisk ocular reflexes) the O_2 consumed by the brain underwent a decrease while if reflexes had disappeared and respiratory gasps or failure and low blood pressure gave indication of deterioration of central nervous functions, cerebral O_2 consumption was increased by the epinephrine. Examples of these responses are shown in table 3.

4. *The effects of convulsant and narcotic drugs.* (a) *Convulsants.* Metrazol, picrotoxin, nikethamide (coramine) and benzedrine were used (table 4).

Metrazol, given intra-arterially or intravenously in about 100 mgm. dosage to 4 responsive animals, led almost immediately to characteristic clonic convulsions involving the face, trunk and limbs. At the same time cerebral blood flow and O_2 uptake were consistently and considerably increased while the A-V oxygen

difference showed relatively slight and inconstant changes. These effects were brief and were typically followed by a prolonged diminution in cerebral functional activity, blood flow and O_2 uptake, with no consistent change in the A-V difference. Given in the same or larger dosage to animals showing no signs of central nervous activity, metrazol was without conspicuous effect on cerebral activity, blood flow, or O_2 uptake. The latter effects therefore depend on the change in functional activity, not on a direct action by the drug on the cerebral blood vessels.

Picrotoxin was given in 3 and 7 mgm. total dosage to 2 animals. It had more marked and more prolonged effects than metrazol. In one experiment (no. 28)

TABLE 3

EXPT. NO.	EXPERIMENTAL CONDITIONS	BLOOD PRESSURE	O_2 CONTENT		CO_2 CONTENT		CEREBRAL			O_2 CONSUMED		CONDITION OF ANIMAL	CORNEAL REFLEX	BRAIN WEIGHT
			Art.	Vein	Art.	Vein	A-VO_2	Blood flow	O_2 uptake	Total	% by brain			
		mm. Hg	*vols. %*		*vols. %*		*vols. %*	*ml./100 g./min.*		*ml./min.*				*grams*
30	Control (nat. resp.)	110	12.5	4.7	52.1	59.6	7.8	53.0	4.13	35	12.3	Excellent	++++	105
	Epineph. infusion	125	10.7	4.6	51.8	58.0	6.1	62.3	3.80	33	12.2	Excellent	++++	
31	Control (nat. resp.)	96	15.4	6.1	46.6	55.0	9.3	45.9	4.26	53	8.7	Excellent	++++	108
	Epineph. infusion	145	13.0	8.4	49.2	52.8	4.6	76.8	3.53	29	13.1	Excellent	++++	
	After epineph.	79	10.6	2.6	45.6	52.3	8.0	51.0	4.08	75	5.9	Excellent	++++	
32	Control (art. resp.)	36	9.6	0.8	67.8	74.6	8.8	21.0	1.85			Bad	0	95
	Epineph. infusion	112	8.5	4.9	66.6	69.0	3.6	85.6	3.08			Fair	+	
8	Control (art. resp.)	32	11.7	2.0	44.3	54.7	9.7	18.9	1.83			Bad	0	95
	Epineph. infusion	94	11.5	6.5	43.9	48.3	5.0	41.5	1.96			Bad	0	
	Epineph. continued	88	9.9	5.7	42.1	46.3	4.2	44.0	1.85			Bad	0	
15	Control (art. resp.)	64	14.2	4.0	81.1	89.3	10.2	17.7	1.80			Bad	0	109
	Epineph. infusion	154	12.5	7.1	80.3	87.3	5.4	44.7	2.40			Bad	0	
	After epineph.	46	8.2	1.8	83.7	90.4	6.4	9.6	0.61			Moribund	0	
12	Control (art. resp.)	77	13.1	8.5	67.7	79.9	4.6	22.7	1.05			Bad	0	90
	Epineph. infusion	164	11.1	5.7	67.8	72.1	5.4	22.4	1.21			Bad	0	
	After epineph.	54	6.9	2.8	52.8	61.1	4.1	10.6	0.43			Moribund	0	

spontaneous breathing was restored and an improvement in blood pressure (probably vasomotor in origin) occurred after the administration of this drug. The increase in cerebral metabolism was due to increase in both A-V difference and blood flow but the preponderance differed in the two animals. The subsequent depressant phase was as evident as with metrazol.

Nikethamide was injected into the carotid stream in 50 to 500 mgm. dosage in 3 monkeys at varying stages of central nervous activity but in only one were there any signs of stimulation of the central nervous system by the drug. Its effects on blood pressure, respiration and cerebral blood flow were peculiar. The first effects after the injection were cessation of respiration, fall in blood pressure, and decrease (amounting in one case to complete standstill for about 20 sec.) in cerebral blood flow. Blood pressure then rose again, spontaneous breathing

TABLE 4

EXPT. NO.	EXPERIMENTAL CONDITIONS	BLOOD PRESSURE	O_2 CONTENT		CO_2 CONTENT		CEREBRAL			O_2 CONSUMED		CONDITION OF ANIMAL	CORNEAL REFLEX	BRAIN WEIGHT
			Art.	Vein	Art.	Vein	A-VO_2	Blood flow	O_2 uptake	Total	% by brain			
		mm. Hg	*vols. %*		*vols. %*		*vols. %*	*ml./100g./min.*		*ml./min.*				*grams*
13	Control (nat. resp.)	114	16.7	8.1	40.9	48.2	8.6	33.5	2.90	29	9	V. good	++	88
	Metrazol 100 mg. i.a.	96	15.4	4.5	31.5	49.2	10.9	47.5	5.17	34	13.8	Convuls.	++	
	Pentothal 5 mg. i.a.	74	16.5	9.8	35.3	36.6	6.7	38.0	2.89	31	9.3	Depressed	0	
	8 min. later	56	14.1	3.0	30.0	42.7	11.1	17.1	1.89	29	5.9	Better	+	
	Transfusion	83	13.2	3.0	34.4	44.1	10.2	31.1	3.16	35	8.0	V. good	++	
14	Control (nat. resp.)	100	13.5	3.5	39.6	49.7	10.0	33.4	3.36	32.6	11.6	Excellent	++++	114
	Pentothal 5 mg. i.a.	100	13.9	2.6	42.9	52.6	11.3	19.0	2.15	26	8.9	Depressed	0	
	Metrazol 100 mg. i.a.	80	12.7	0.6	37.3	46.9	12.1	28.8	3.50	25	16	Convuls.	++	
	15 min. later	64	14.1	2.3	45.7	54.7	11.8	17.3	2.04	30	7.7	Depressed	0	
	30 min. after met.	110	11.7	2.4	43.3	53.1	9.3	31.4	2.92	36	9.2	V. good	++	
10	Control (nat. resp.)	100	12.9	5.0	58.3	68.3	7.9	29.2	2.30			Poor	0	85
	Metrazol 100 mg. i.a.	136	11.1	4.2	56.7	69.4	6.9	59.8	4.13			Convuls.	++	
	15 min. later	100	5.7	1.6	66.3	71.0	4.1	20.6	0.85			Bad	0	
18	Control (nat. resp.)	94	15.4	3.4	14.4	27.6	12.0	22.6	2.71	29.4	10.9	Fair	0	102
	Metrazol 100 mg. i.a.	124	14.6	2.9	15.7	27.0	11.7	30.1	3.53	33.3	9.3	Convuls.	++	
8	Control (art. resp.)	88	9.9	5.7	42.1	46.3	4.2	44.0	1.85			Bad	0	95
	Metrazol 100 mg. i.a.	80	10.1	7.1	39.8	44.1	3.0	65.7	1.97			Sl. convul.	++	
32	Control (art. resp.)	112	8.5	4.9	66.6	69.0	3.6	85.6	3.08			Bad	0	95
	Metrazol 200 mg. i.v.	80	5.3	0.6	61.9	68.4	4.7	62.2	2.92			No convul.	0	
	Coramine 500 mg. i.v.	80	6.3	0.8	36.4	51.7	5.5	66.4	3.65			Convuls.	++	
	17 min. later	40	5.8	0.3	40.4	52.8	5.5	27.4	1.51			Bad	0	
	12 min. later	34	5.7	0.3	45.9	53.3	5.4	24.6	1.32			Moribund	0	
22	Control (nat. resp.)	128	8.5	2.4	62.0	67.6	6.1	32.1	1.95	42	4.0	Poor	0	
	Coramine 50 mg. i.a.	70	8.8	2.5	65.4		6.3	26.6	1.67	40	3.8	No convul.	0	
	9 min. later	122	8.8	2.6	59.7	67.0	6.2	29.3	1.82	37	4.3	Poor	0	
	Pentothal 5 mg. i.a.	52	9.1	2.7	71.3	74.8	6.4	13.5	0.87			Bad	0	
24	Control (nat. resp.)	106	11.1	4.1	51.5	60.0	7.1	54.2	3.85	36	6.2	Good	+	85
	Coramine 50 mg. i.a.	130	9.7	4.5	49.9	56.8	5.2	64.4	3.24	41	6.8	No convul.	+	
	4 min. later	106	8.4	0.7	55.6	60.5	7.7	28.7	2.21	31	6.1	Poor	0	
27	Control (nat. resp.)	102	13.0	4.2	40.2	50.9	8.8	49.7	4.37	35	9.4	Excellent	++++	75
	Picrotoxin 3 mg. i.a.	116	12.5	3.6	41.7	50.7	8.9	57.0	5.06	40	9.5	Convuls.	++++	
	19 min. later	70	12.7	3.7	51.7	62.8	9.0	41.0	3.68			Poor	0	
	8 min. later	102	12.8	2.0	53.8	65.0	10.8	60.1	6.50			Twitching	++	
	8 min. later	80	12.5	2.1	49.4	62.5	10.4	51.5	5.35			Convuls.	++	
	42 min. later	124	11.8	2.0	44.8	51.8	9.8	48.6	4.77	36	10	Excellent	++++	
	Pentothal 2.5 mg. i.a.	116	11.4	3.1	46.2	54.3	8.3	45.9	3.81	38	7.6	Depressed	±	
	15 min. later	124	10.5	2.2	46.5	52.5	8.7	50.3	4.37	38	8.7	Excellent	++++	
28	Control (art. resp.)	83	14.5	7.5	68.4	74.9	7.0	33.8	2.36			Bad	0	88
	Picrotoxin 7 mg. i.a.	104	14.2	2.0	70.2	84.1	12.2	34.6	4.22	36	10.2	Good	+	
	25 min. later	102	14.1	1.9	65.5	78.4	12.2	29.2	3.57	36	8.6	Good	+	
	Benzedrine 1 mg. i.a.	106	12.7	3.1	65.0	75.8	9.6	27.0	2.59	36	6.4	Poor	0	
	18 min. later	106	11.8	2.6	67.1	74.9	9.2	30.5	2.80	52	4.8	Fair	0	
2	Control (art. resp.)	80	9.5	3.5	34.4	38.3	6.0	25.2	1.51			Bad	0	95
	Benzedrine 5 mg. i.v.	94	10.5	3.3	35.4	40.1	7.2	29.0	2.1			Bad	0	
	5 min. later	86	6.7	2.0	36.5	40.7	4.7	25.4	1.2			Bad	0	
5	Control (art. resp.)	64	12.7	4.6	38.4	47.5	8.1	36.4	2.95			Bad	0	90
	Benzed. 5 mg. i.v.	70	11.0	1.7	36.9	47.1	9.3	20.8	1.94			Bad	0	

returned, cerebral blood flow came back to or above its control level, and the convulsant effects of the drug now appeared for the first time.

Benzedrine was given in a number of experiments to test its effects on cerebral blood flow but only in 3 of these was there a sufficiently steady state to justify the collection of blood samples. Cerebral metabolism was somewhat decreased in 2 of the cases, increased in the third. Cerebral blood flow behaved like cerebral metabolism. Blood pressure rose in all 3 but there were no manifestations of stimulation of the central nervous system.

Representative examples of these results are shown in table 4. Cerebral O_2 uptake and blood flow were invariably increased during the convulsant period by metrazol, nikethamide and picrotoxin, and this therefore appears to be the characteristic pattern of the action of such drugs. The behavior of the A-V oxygen difference was so inconstant as to indicate that factors at present unidentified are involved here. The absence of effect on either metabolism or blood flow when no convulsions were seen indicates that the increase in blood flow is related to the convulsant action and is not simply due to dilatation of cerebral vessels by the drug itself. The convulsant period was invariably followed by depression of respiration, ocular reflexes, cerebral blood flow, and cerebral O_2 uptake, the latter reaching approximately the level attained by a distinctly narcotic dose of pentothal (table 4). The time relations of these various events differed considerably among the three convulsants. With metrazol the maximum effects on cerebral activity, blood flow and metabolism were seen almost immediately after the injection and the stimulant phase lasted only a minute or two. With picrotoxin the effects came on more gradually but were much more prolonged; in one case (expt. 27) the initial stimulant phase was followed by a period of depression and this by a return of the convulsant effects of the original dose of the drug, indicating that the initial stimulant action was terminated by deterioration in the ability of the brain to react, not by excretion or detoxification of the drug. With nikethamide there was an initial depressant action on cerebral functions and blood flow that was lacking with the other two. The stimulant phase came on after this (if at all) and was accompanied by the same type of increase in cerebral blood flow and metabolism that was seen with the others. The subsequent depression of cerebral functional activity, metabolism and blood flow seemed to depend more on the vigor of the convulsant response than on the drug used and was of about the same order of intensity and duration with all 3 agents.

Benzedrine, in the dosage employed and under the existing experimental conditions, gave no sign of an ability to stimulate the brain in a manner comparable with the convulsants. The effects obtained by it could be ascribed to its sympathomimetic actions associated with a mild constriction of cerebral blood vessels; the latter action would be a unique and unfavorable concomitant of stimulation of brain cells, but it is possible that if such stimulation occurred the relatively mild cerebral constriction would give way to vasodilatation. On the other hand, if stimulation did not occur the cerebral vasoconstrictor action would be distinctly disadvantageous.

(b) *Depressants.* Pentothal sodium is the only agent with which enough ob-

servations were made to warrant definite statements. It was chosen because of the brevity and complete reversibility of its effects When given into the carotid stream in suitable dosage (2–5 mgm. total) to animals with active respiration and reflexes, it caused distinct depression of rate and depth of breathing and disappearance of the ocular reflexes, but artificial pulmonary ventilation usually was not required and systemic blood pressure was not significantly altered. At the height of this effect cerebral O_2 uptake and blood flow were both reduced consistently and considerably while the A-V oxygen difference showed no consistent change. Pentothal was also given to counteract the convulsant effect of metrazol; the effects on cerebral blood flow, A-V difference and metabolism were of the same type as those just described though the true effect of the depressant drug is not as clear because of the spontaneous tendency toward a temporary depressant phase in these functions following the convulsant period. Examples of these effects are shown in table 4.

5. *Correlations among cerebral blood flow, A-V oxygen difference and oxygen uptake.* (a) *Between cerebral blood flow and A-V oxygen difference.* Our data bearing on this are shown in figure 2. In this (as well as in figs. 3, 4 and 5) each point represents one set of blood samples with its corresponding observations and all of the data are included without selection. Although in a given experiment there may be good correlation between these two factors within certain limits (tables 2 and 3), when all the findings in all the experiments are viewed together this relationship has a correlation coefficient of -0.18, which is the poorest of the three possible ones among these three variables.

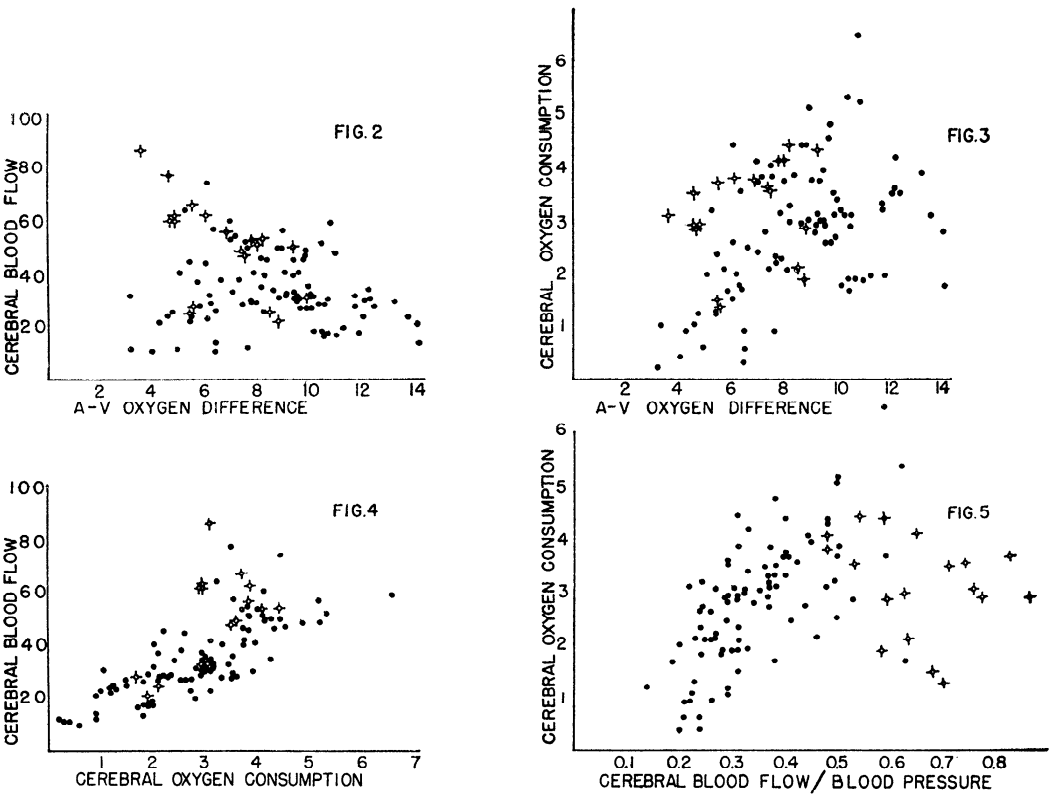
(b) *Between cerebral oxygen consumption and cerebral A-V oxygen difference.* Our findings on this are plotted in figure 3. The correlation coefficient here is 0.54, which is considerably better than (a).

(c) *Between cerebral oxygen consumption and cerebral blood flow* (fig. 4). The coefficient of correlation here is 0.67, which is the best of the three. This suggests that the cerebral circulation automatically adjusts itself to the existing metabolic requirements of the brain—a suggestion that is borne out by figure 5, which shows the relationship between cerebral metabolic activity and cerebral blood flow corrected for differences in blood pressure, so as to disclose changes in cerebral vascular tone. The coefficient of correlation for the findings in rhesus monkeys is 0.59, which indicates a significant relationship. The data from spider monkeys fall completely out of line, the blood flow per unit of tissue and blood pressure tending to be considerably greater than in the rhesus monkeys. This is already evident in table 1, in which numbers 30 and 31 were ateles and the others were rhesus. A species difference is strongly suggested but the data are insufficient to establish a separate set of correlations. No corresponding differences are evident in the other relationships (figs. 2, 3 and 4).

6. *Other observations.* (a) *The effects of insulin.* Only one of three experiments with this drug was satisfactory. The results are shown in table 5. The absence of any striking change in cerebral O_2 uptake is noteworthy. The animal's condition was excellent and spontaneous muscular movements, which were present before insulin, increased during the hypoglycemic period. The decrease in O_2

uptake following glucose injection was probably of the same nature as that seen with epinephrine in "normal" animals (table 3). Unfortunately this animal showed no trace of hypoglycemic shock in spite of a total of 20 units of insulin and these observations therefore apply only to the stage of moderate ("therapeutic") insulin action.

(b) *The effects of changes in arterial pO_2 and pCO_2.* Observations along these lines were incidental and were usually made after the planned experiment on



Figs. 2, 3, 4 and 5. The values for cerebral blood flow and cerebral O_2 consumption are plotted in terms of milliliters per 100 grams wet brain weight per minute, those for A-V oxygen difference in volumes per cent. In figure 5 the observed blood flow values have all been divided by the blood pressure at the time of measurement. The observations on spider monkeys are shown as stars; all the others were made on rhesus animals.

cerebral metabolism had been made. The results confirmed those of our earlier work on monkeys (6) in all essential respects: inhalation of 10 per cent CO_2 in O_2 had little or no effect on cerebral blood flow unless blood pressure underwent a change, which the flow followed apparently passively. Anoxemia was associated with an increase in flow provided that cerebral activity was present, but not otherwise. Induced hyperventilation with O_2 did not decrease the blood flow unless blood pressure fell. The extent of the changes in arterial gas tensions was not determined.

(c) *The effects of breathing against pressure.* These observations, like the preceding, were incidental and were made on animals that were still in good condition after the planned metabolism experiment had been made. The pressure was hydrostatic and was applied to the outlet of the expiratory valve. It was therefore continuous throughout the respiratory cycle. The inspired gas was 100 per cent O_2. Satisfactory tests were made on 5 rhesus monkeys. The results are adequately summarized by stating that cerebral blood flow did not change until the blood pressure did, when both changed in the same direction. Pressures up to 3 inches of water only slowed breathing somewhat without affecting blood pres-

TABLE 5

EXPT. NO.	EXPERIMENTAL CONDITIONS	B-P	O_2 CONTENT		CO_2 CONTENT		CEREBRAL			O^2 CONSUMED		BLOOD SUGAR	CONDITION	C.R.	WT. OF BRAIN
			Art.	Vein	Art.	Vein	A-VO_2	Blood flow	O_2 uptake	Total	% by brain				
			vols. %		*vols. %*		*vols. %*	*ml./100g./min.*		*ml./min.*		*mgm./%*			
31	Control	79	10.6	2.6	45.6	52.3	8.0	51.0	4.08	75	5.9	159	Excel.	++++	108
	10′ after insulin-20 u	66	9.9	2.4	46.4	54.2	7.5	47.2	3.54	70	5.0	98	Excel.	++++	
	1 hr. after insulin	66	9.1	1.7	47.2	54.5	7.4	49.3	3.64	75	4.9	41	Excel.	++++	
	After glucose-5 g.	69	6.0	1.2	49.3	52.7	4.8	60.3	2.89	76	3.8	197	Excel.	++++	

TABLE 6

	FINDINGS IN MONKEYS' BRAINS ML./100 G./MIN.		Q_{O_2}	CORRESPONDING VALUE FOR 1400 G. HUMAN BRAIN (ML./MIN.)	
	O_2 uptake	Blood flow		O_2 uptake	Blood flow
"Normal" mean	3.7	47	11.1	52	660
"Normal" minimum	2.5	33	7.5	35	460
"Normal" maximum	4.5	74	13.5	63	1040
Highest of all (picrotoxin convulsion)	6.5	91	19.5	91	1275
Lowest with recovery (hemorrhage)	1.85	13.2	5.6	26	185
Lowest of all (moribund)	0.39	5	1.17	6	70

sure or cerebral blood flow but at 4 inches respiratory difficulty and fall in blood pressure were seen. None of the animals were able to breathe against 8 inches: the chest assumed the inspiratory position and the expiratory muscles apparently were not strong enough to expel gas against this pressure; blood pressure fell sharply and cerebral blood flow did likewise.

These findings indicate that the rise in systemic venous pressure associated with high intrapulmonary pressure does not lead to diminution in arterial inflow to the brain, over the range of pressure that can be borne by the respiratory and cardiovascular apparatus.

(d) *The effect of minimal movement of blood in the cerebral circulation.* Because of their possible bearing on the problem of resuscitation we wish to record here a few observations which were made possible by the unique conditions of these

experiments. In a number of cases the animal was still in good condition (active reflexes, normal type of breathing) when terminal exsanguination was begun. The bleeding was rapidly followed by gasping respiration, then by complete failure of respiration and circulation. We discovered accidentally in one such experiment that if blood was sucked out of the venous catheters into a syringe, respiratory activity could be restored and maintained for some time, and this was confirmed in subsequent similar experiments. Apparently movement of blood through the cerebral vessels led to distinct improvement in the state of the respiratory center even though the pressure in the vessles was zero (atmospheric) or below. This course of events was seen only in animals previously in good condition, with respiratory failure resulting from rapid exsanguination, and with a minimum lapse of time between the failure and the withdrawal of venous blood—a situation having its counterpart in ventricular fibrillation associated with electric shock. If the two are really comparable, the prompt application of manual artificial respiration should help to prevent total and permanent failure of the respiratory center in cases of the latter type.

(e) *The fraction of the total oxygen uptake accounted for by the brain.* In the 9 "normal" animals in which pertinent data were obtained, the average for this was almost exactly 10 per cent (table 1). The changes that took place under changing conditions are shown in the subsequent tables. It should be recalled that most of the injections of convulsants and all of the injections of pentothal were made intra-arterially so that only the brain was exposed to an effective concentration of these drugs.

(f) *The partition of the total carotid flow between the internal and external carotids.* This was studied in one spider monkey in which the basilar artery was ligated and the flow started through the meter before the external carotids were tied. The flow was 52 ml. per minute at a blood pressure of 78 mm. Hg. After the external carotids were tied the flow was 37 ml. per minute at a pressure of 80 mm. In this animal about 70 per cent of the total common carotid stream was therefore carried by the internal carotids.

DISCUSSION. The importance of the four main physiological variables dealt with in these experiments—cerebral functional activity, cerebral metabolic activity, cerebral blood flow and cerebral arteriovenous oxygen difference—is attested by the existence of a large mass of literature dealing with each. The mutual relationships among them have been the basis for considerable speculation, but as far as we are aware this is the first time that direct experimental evidence bearing upon these relationships has been obtained under conditions approximating the normal. While we cannot guarantee that these findings are applicable to animals other than the monkey, or even to that animal in the intact, unanesthetized state, we believe that they deserve full discussion because they afford a glimpse of these interrelationships under conditions which, although they cannot be called entirely normal, certainly were not incompatible with cerebral functional activity. To facilitate this discussion we have summarized some of the most important findings in table 6.

The normal cerebral Q_{O_2}. In view of the depression of cerebral O_2 uptake by

narcosis (table 4) it seems proper to regard the highest of our "normal" values as our closest approximation to the situation in the intact, unanesthetized monkey. This (4.5 ml. per 100 grams of wet brain per min.) gives a Q_{O_2} of 13.5, which comes directly in the midst of the findings for slices of guinea pig or rat cerebral cortex *in vitro* (p. 33 above). The natural conclusion would be that the *in vitro* technic gives values that closely approximate the O_2 uptake by the living mammalian brain *in vivo*. Before this conclusion can be accepted, however, two facts must be taken into consideration. One is that narcosis was present throughout the present experiments; even the highest "normal" Q_{O_2} should have been somewhat depressed by this. The other is that these *in vivo* values are for the entire brain whereas the *in vitro* ones which they most closely resemble are for cortex alone. If the uptake of the cortex in our monkeys was 2 to 4 times that of subcortical areas, as *in vitro* studies have indicated it to be in other animals (15), it follows either that the metabolic rate of the monkey's cortex must have been much higher than our figure, or that practically all of the measured blood flow and O_2 uptake were accounted for by the cortex. The latter seems improbable in view of the active state of the ocular reflexes and of the respiratory and vasomotor centers. The former alternative therefore is preferable, but in that case the agreement between our Q_{O_2} and that of cortical slices probably was purely fortuitous.

It is noteworthy that even the highest of our Q_{O_2} values is considerably below any of those previously reported for the mammalian brain *in vivo* (p. 33). This discrepancy almost certainly is due to overestimation of the blood flow factor in the earlier experiments because of escape of blood through anastomoses with the extracranial circulation. This subject is considered in detail elsewhere (20) and needs no further discussion here. The "physiological" range of the Q_{O_2} (i.e., the highest attainable with a convulsant and the lowest compatible with complete recovery) turns out to be from nearly double to about half the resting "normal" value (p. 36 above). Lower values than this were encountered but since neither cerebral functions nor cerebral O_2 uptake could then be restored such levels probably correspond with irreversible (pathological) changes in nerve cell function.

The magnitude of cerebral blood flow. Table 6 gives a fair representation of the range of cerebral blood flow in these animals under conditions compatible with cerebral activity, as well as the corresponding values for an adult human brain of average size. It is noteworthy that the mean and maximum blood flow values encountered in these monkeys were distinctly below the corresponding figures (60 and 113 ml. per 100 grams per min. respectively) in our earlier work (6). Part of the difference may be due to the routine use in these experiments of inhalation of 100 per cent oxygen, which tends to reduce cerebral blood flow in the monkey (6). Part may be attributable to the fact that the monkeys used in the present study were on the whole smaller and less resistant to the experimental procedures, perhaps because of longer captivity. We have not succeeded in a number of recent attempts at checking the accuracy of our earlier correction factor of 70 per cent for the reduction in cerebral blood flow upon closure of the basilar artery (6). The basilar artery of the spider monkey is so much smaller

than that of the rhesus that the same correction certainly should not be applied to both. The figures presented throughout this paper are those actually obtained and are uncorrected.

The significance of changes in the cerebral A-V oxygen difference. We have pointed out in section 5 that although in a given animal and within certain limits a change in the A-V difference can be correlated with a simultaneous change in either cerebral blood flow or cerebral metabolism, there is poor correlation between the A-V difference and either of these factors when the data are viewed together. The best of the three possible correlations among these variables is that between cerebral metabolism and cerebral blood flow, which, as already pointed out (p. 42), strongly suggests an effective adjustment of cerebral blood flow to the momentary metabolic requirements of the brain. An alternative explanation would be that cerebral blood flow in some manner determines cerebral metabolism. While this undoubtedly is true when a subnormal flow has become the limiting factor on cerebral metabolism, it does not appear to hold when the flow is increased above the optimal level (table 3) or reduced only enough to permit adequate compensation by increasing the A-V difference (table 2). We therefore prefer the former interpretation, which is in agreement with all our experimental findings. But if there is a completely effective mechanism for automatically regulating the blood supply of the brain in accordance with its metabolic requirements, it follows that the A-V difference of oxygen (or anything else) should be kept constant, by suitable changes in cerebral blood flow, in the face of considerable changes in cerebral metabolic activity. Any changes in the A-V difference would signify that the mechanism had not functioned perfectly but there would be no clue as to whether the primary factor was a change in blood flow or in metabolism. Our results with primary changes in blood flow (tables 2 and 3) and metabolism (table 4) show the impossibility of drawing valid deductions of this character from changes in the A-V difference alone. This question is considered in greater detail elsewhere (20).

The relation of cerebral functional activity to cerebral metabolism. One of the most striking and consistent findings in these experiments is the direct relation between cerebral functional activity (manifested in respiratory and vasomotor control, ocular reflexes, muscular movements) and cerebral O_2 consumption. Our findings therefore confirm those already obtained in other animals with other methods (1) (18), and justify the addition of the mammalian brain to the list of tissues in which an increase in functional activity has been proved to be associated with an increase in its call for oxygen (2). The causal nature of this relationship is not disclosed by our experiments but it seems most probable that each may be the cause or each the effect, depending on circumstances. When the primary factor is reduction in the supply of O_2 (as in our hemorrhage experiments, but presumably also in any other type of cerebral anoxia) the decrease in functional activity must be the result of the anoxic interference with cerebral cellular activity. Under such circumstances restoration of the O_2 supply to the brain will lead to return of functional activity and O_2 uptake, provided that the anoxic derangement has not progressed far enough to be partly or completely irrever-

sible. According to our results with hemorrhage the degree of reduction in cerebral O_2 uptake that can be recovered from promptly and completely is about half the resting "normal".

These statements imply that reduction in the cerebral O_2 supply will necessarily lead to a diminution, and increase of the O_2 supply to an increase, in cerebral O_2 uptake and functional activity. While this is generally true, we have encountered exceptions in both directions and have some data by which they may be understood. Reduction in cerebral O_2 supply produced by gradually progressing cerebral anemia (hemorrhage) invariably led to an increase in the A-V oxygen difference. As long as this was sufficient to compensate for the diminution in blood flow and to prevent a significant decrease in cerebral O_2 uptake, little or no alteration in cerebral functional activity could be detected, but once the limit of this type of compensation had been reached and the O_2 uptake began to fall, deterioration rapidly ensued (table 2). At the opposite extreme we have found two sets of circumstances under which an increase in cerebral blood flow did not lead to an increase in cerebral O_2 uptake. One was the intravenous infusion of epinephrine in animals whose cerebral functions, blood flow and metabolism were optimal at the time (table 3, expts. 30 and 31); the decrease in O_2 uptake actually observed is probably referable to an uneven effect on the A-V oxygen difference in different parts of the brain, that in the functionally silent areas (where the A-V difference would be decreased markedly by an increase in blood flow because the demand for O_2 must be relatively small) apparently assuming dominance over that in the active regions (where the decrease in the A-V difference presumably would be just enough to compensate for the increased flow). This explanation was already put forward to account for a similar effect from CO_2 inhalation in dogs whose total cerebral blood flow was maintained constant (18). It seems preferable to the alternative possibility that epinephrine depresses cerebral O_2 consumption, not only because of the well-known capacity of this drug to produce central nervous excitation in man, but also because of the increase in cerebral O_2 uptake produced by it when cerebral functions were depressed (table 3). Increase in cerebral blood flow also failed to increase cerebral O_2 consumption in some experiments when, following hemorrhage, transfusion was delayed until respiratory failure had occurred (table 2, expt. 25). This confirms, in another animal and with other methods, the existence of a state previously designated as the "reversal", characterized by prolonged or permanent diminution in the ability of brain cells to take oxygen out of the blood and associated with a corresponding loss of cerebral functional activity (18).

Thus a decrease in cerebral blood flow need not lead to a diminution in cerebral O_2 uptake if the former is small (and brief) enough to be compensated fully by increase in the A-V oxygen difference; an increase in flow does not cause an increase in O_2 uptake if the existing state of cerebral function, flow and metabolism is already optimal, or if cerebral anoxia has been sufficiently severe and prolonged to interfere with the ability of brain cells to utilize oxygen (the "reversal" (18)). With these exceptions, and within these limits, cerebral O_2 consumption varies directly with cerebral blood flow when the change in the latter is primary.

When the change in cerebral O_2 uptake is brought about by a convulsant or narcotic drug there is also a consensual change in cerebral functional activity and blood flow, but here the alteration in cerebral blood flow probably is the result of a primary change in cerebral metabolism. Characteristic features such as dependence of the increase in O_2 uptake and blood flow on increased functional activity, occurrence of a depressed phase in all three after a period of convulsions, irregular and uncertain behavior of the A-V oxygen difference, and individual variations among the convulsant drugs used, have already been presented (section 4a) and need not be repeated here. The occurrence of prolonged or permanent depression of cerebral functions and metabolism (the "reversal") following various stimulant procedures was demonstrated in earlier work (18). Our present findings indicate that the O_2 supply to the brain need not be reduced at the time of the stimulation for this to occur, as then seemed to be the case, for in the present experiments anoxemia did not occur and cerebral blood flow was invariably increased during the convulsions. Apparently an increase in cerebral O_2 consumption to something less than double its resting level by means of a convulsant drug (table 4) will be followed by a state characterized by depressed cerebral functional activity, O_2 uptake and blood flow, which is identical with the effect of cerebral anoxia due to hemorrhage. It is interesting that nearly doubling the O_2 demand had the same type of reversible depressant effect as halving the O_2 supply. Since a prolonged depression of cerebral functions follows the convulsions produced by sodium cyanide, which apparently owes all these effects to reflexes from the carotid and aortic bodies (since they are lacking when the drug is given after denervation of these structures (19)), it is evident that the phenomena of post-convulsant depression do not depend entirely on a direct diphasic action by the convulsant drug on nerve cells, but are more probably related to the increased cerebral metabolic activity associated with the convulsions. Thus anoxia of the brain, with all its manifestations, can be produced quite as well by increasing the O_2 requirement to exceed the available O_2 supply, as by reducing the supply without altering the requirement. We have also tried convulsant drugs when ocular reflexes had disappeared and slow or absent breathing and low blood pressure gave indication of cerebral depression. With a single exception, the drugs uniformly failed to improve cerebral functional activity or O_2 uptake under such circumstances. The exception was an experiment (table 4, expt. 28) in which, following a large total dose of picrotoxin, blood pressure rose, cerebral A-V oxygen difference increased, ocular reflexes returned and spontaneous breathing was resumed; no convulsions occurred, but the improvement of blood pressure, respiration and ocular reflexes, associated with an increase in the A-V oxygen difference (without significant change in cerebral blood flow) shows that a direct stimulation of nerve cells can be produced by a convulsant drug in the presence of depression. The margin between this and the onset of convulsions, leading to a more severe post-convulsant depression, may however be a narrow one. Furthermore, this instance of stimulation was altogether exceptional, probably because the factor responsible for loss of cerebral functions usually was anoxia, the effects of which would be readily removed by an increase

in cerebral blood flow (as by epinephrine—table 3) if the "reversal" had not occurred, and if it had, convulsant drugs could scarcely remedy the cellular disorganization which this implies. Our present impression is that convulsant drugs are not likely to prove beneficial in the presence of cerebral anoxia unless they lead to a rise in blood pressure. They may shorten the period of recovery following a depressant drug such as pentothal, but in our animals the stimulant effect of the analeptic was followed by a depression in cerebral O_2 consumption to a level at least as low as that produced by the depressant in the first place and the total recovery period seemed to have been prolonged rather than shortened. This is in accord with the finding by Mousel and Essex (13) that analeptic drugs delay the recovery from pentothal in dogs, cats and rabbits.

The effects of pentothal on cerebral functions, O_2 uptake and blood flow were the precise opposite of those of the convulsants except for the subsequent swing in the opposite direction, which did not occur with pentothal (table 4). The ability of this drug to produce a distinct diminution in cerebral O_2 uptake when it is given in moderately effective dosage thus is demonstrated and the findings of Quastel and Wheatley (15) are confirmed. Recovery from the depressant effects of pentothal took place more rapidly than was the case following a comparable diminution in cerebral O_2 uptake produced by hemorrhage or convulsant drugs. This suggests that the effect of pentothal is the more benign and is exerted in a different manner. If it is true that the depression following a convulsant is basically the same as that following hemorrhage and is due to cerebral anoxia created by increasing the O_2 requirement more than the O_2 supply, it becomes probable that a decrease in cerebral O_2 uptake produced by a drug such as pentothal might increase the ability of the brain to withstand the effects of anoxia however produced. The margin of safety may be relatively narrow and the gain would be bought at the price of marked if not total interference with the capacity to perform mental and physical tasks, but for situations in which avoidance of severe cerebral damage from anoxia is a critical consideration and consciousness on the part of the subject is not essential this procedure might be beneficial. Examples would be severe poisoning by CO, marked anoxemia from respiratory obstruction, pulmonary edema, inhalation of high concentrations of N_2O or C_2H_4 for surgical operations, or cerebral anemia due to low blood pressure or high cerebrospinal pressure.

The intrinsic control of the cerebral circulation. Figure 5 indicates not only that cerebral blood flow is automatically adjusted to the metabolic requirements of the brain, but also that the adjustment does not depend on passive effects from changes in systemic blood pressure. As to the nature of this adjustment, we can only confirm our earlier finding (6) that total cerebral blood flow in the monkey is affected more markedly and consistently by changes in pO_2 than by changes in pCO_2 in the arterial blood. As between the two principal metabolic gases O_2 therefore appears to be the more likely mediator of an intrinsic adjustment of this type, but it is much more probable that the control is achieved by summation of all of the vasodilator products of metabolism (changes in pCO_2 and pO_2, acid-base adjustments, local changes in temperature, perhaps liberation of K, histamine,

acetyl choline, etc.) and that no one agency should be singled out as being solely responsible. We have noted elsewhere (20) that the current widespread inclination toward CO_2 as the main or sole factor depends on the accidental circumstance that most of the recent work in this field has been done on the circulation in the cerebral cortex of the cat. The situation there is not necessarily representative of that in the cortex of other animals, nor the situation in the cortex of any animal of that in other parts of its brain.

The vasomotor response in these experiments showed parallelism with the respiratory. In the hemorrhage experiments we were able to predict which animals would make a complete recovery by observing the behavior of their blood pressure and cerebral blood flow after the transfusion, for those in which these came back to or above the control level recovered, and those in which they did not do so failed to recover. In the single case of restoration of cerebral functions by an analeptic drug recovery of vasomotor tone began before ocular reflexes or spontaneous breathing returned. Our findings indicate that the vasomotor center is one of the most susceptible of intracranial contents to anoxic derangement, and not the most resistant, as the hypertension associated with acute elevation of intracranial pressure suggests (23). This question also is considered elsewhere (20). At present it seems quite likely that the onset of sudden collapse at high altitude is due to absence of the vasomotor response, as has already been suggested (21). If this is true for this type of cerebral anoxia it should be equally true for others, and the vasomotor center then may play the decisive rôle in determining whether cerebral functions will or will not be maintained. Loss of consciousness if the subject is in the erect position then may be viewed as a safety reaction, intended to safeguard venous return to the heart (21). If it is also associated with a diminution in cerebral O_2 uptake (which should be the case in unconsciousness of any cause), this would be an added safety measure to protect the brain from O_2 utilization in excess of the available O_2 supplv

SUMMARY

1. Cerebral O_2 metabolism has been measured *in vivo* in lightly anesthetized monkeys by measuring cerebral blood flow directly while samples of cerebral venous and arterial blood were collected for subsequent analysis. Important findings:

	MONKEY'S BRAIN ML./100 G./MIN.		Q_{O_2}	CORRESPONDING VALUE FOR 1400 G. HUMAN BRAIN (ML./MIN.)	
	O_2 uptake	Blood flow		O_2 uptake	Blood flow
"Normal" mean	3.7	47	11.1	52	660
"Normal" minimum	2.5	33	7.5	35	460
"Normal" maximum	4.5	74	13.5	63	1040
Highest of all (picrotoxin convulsion)	6.5	91	19.5	91	1275
Lowest with recovery (hemorrhage)	1.85	13.2	5.6	26	185
Lowest of all (moribund)	0.39	5	1.17	6	70

2. Cerebral O_2 uptake invariably changed in the same direction as cerebral functional activity (judged by muscular movements, ocular reflexes, character of respiration, level of blood pressure) whether the latter underwent spontaneous changes or was altered by changes in cerebral blood flow (hemorrhage transfusion, epinephrine infusion) or by convulsant (metrazol, picrotoxin, nikethamide) or depressant (pentothal) drugs. The "physiological" range of cerebral O_2 uptake was from about half to nearly double the resting "normal" value. Convulsions were followed by a period of depressed O_2 uptake of the same order as that produced by a deeply narcotic dose of pentothal.

3. Of the 3 possible correlations among (a) A-V oxygen difference, (b) blood flow and (c) O_2 uptake of the brain, that between (a) and (b) was poorest, that between (a) and (c) somewhat better, but that between (b) and (c) by far the best. This is believed to indicate that the tone of cerebral blood vessels is automatically adjusted to the metabolic requirements of the brain. Changes in A-V oxygen difference then signify only that this adjustment has been imperfect and do not justify conclusions as to changes either in cerebral blood flow or metabolism. Previous work indicating that the cerebral circulation of the monkey is affected more consistently and strongly by changes in pO_2 than in pCO_2 has been confirmed; as between these, pO_2 is the more likely to be responsible for the intrinsic control, though it is probable that any or all of the vasodilator products of metabolism (CO_2, anoxia, acid, heat, K, acetyl choline, histamine, etc.) may play a part.

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